

DiaCausal datasets: what we use, why synthetic, and what real Indian data exists

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

For the mid-sem slides and the viva. Every number below comes from a file in the repo or from a script run on 28 Sept 2026; the pictures are in `docs/midsem/data/`.

1. The datasets in one table (slide-ready)

#	Dataset	Real or synthetic	Size	What it is for	File	Picture
1	DiaCausal synthetic cohort	Synthetic (India-calibrated)	5,000 patients per cohort, 20 cohorts in the benchmark	Train and grade the causal engine (the true answer is known)	made by <code>diacausal_engine/cohort.py</code> from <code>data/params.yaml</code>	<code>01_synthetic_cohort.png</code> , <code>02_synthetic_hidden_truth.png</code>
2	Safety rules	Real (drug labels, KDIGO)	10 rules	Remove unsafe options before any estimate	<code>data/rules.csv</code>	<code>03_safety_rules.png</code>
3	Generator parameters	Each number labelled CITED / ASSUMED-DIRECTIONAL / TEAM-SET	117 numbers	Say how the synthetic patients are made	<code>data/params.yaml</code>	<code>04_generator_parameters.png</code>
4	Evidence sources	Real documents (WHO 2018, 7 US FDA safety notices)	26 sources listed, 8 texts in the search after the licence check	RAG evidence and explanations	<code>RAG/sources.csv</code> , <code>diacausal_rag/corpus/</code>	<code>05_rag_sources.png</code>
5	Evidence test set	Written by the team	60 questions (45 answerable, 10 out of scope, 5 dose)	Measure the evidence search	<code>eval/rag_gold.csv</code>	<code>06_rag_gold_questions.png</code>
6	NMB-2017 (new)	Real Indian data (CC0)	7,496 adults, 1,986 with self-declared diabetes	Check that the synthetic patients look like real Indian patients	<code>data/reference/nmb2017.csv</code>	<code>07_real_indian_nmb2017.png</code> , <code>real_vs_synthetic.png</code>

Regenerate everything: `python scripts/fetch_nmb2017.py`, `python scripts/compare_cohort.py`, `python scripts/dataset_previews.py` (the last needs `CHROMIUM_PATH` or Playwright's own browser).

2. The synthetic cohort, column by column

One row is one adult with type 2 diabetes who is already on metformin and whose HbA1c is still at or above 7%. The generator gives each one an add-on drug the way doctors tend to, then records what happened by 6 months.

Column	Meaning	Unit / values
age	Age	years (30–85)
female	Sex	1 = female (45%)
duration_years	Years since diagnosis	years
hba1c	HbA1c at the start	% (7.0–13.0)
egfr	Kidney function; falls with age	mL/min/1.73m ² (30–120)
bmi	Body-mass index; Asian-Indian cut-offs (overweight ≥ 23, obese ≥ 25)	kg/m ²
ascvd, hf	Heart disease, heart failure	1 = yes
hypo_history, dka_history, pancreatitis_history	Past low sugar, ketoacidosis, pancreatitis	1 = yes
low_income	Cost matters	1 = yes
ckd	eGFR < 60 (KDIGO)	1 = yes
treatment	Add-on drug given	SGLT2i / DPP4i / SU
y	Outcome: HbA1c change at 6 months	percentage points
weight_change_6m	Weight change at 6 months	kg
hypo_6m	Any low-sugar episode by 6 months	1 = yes
y_true_*, e_true_* ...	Hidden truth (never shown to the estimators)	—

How it is generated (5 steps, `diacausal_engine/cohort.py`)

1. **Draw the patients.** Age, sex, years with diabetes, HbA1c, eGFR (lower in older patients), BMI and the history flags, from the distributions in `params.yaml`.
2. **Give each one a drug the way doctors tend to.** A formula makes the choice depend on the patient (for example higher HbA1c and BMI → more SGLT2i; low eGFR → less SGLT2i). This is *confounding by indication*, the same bias hospital records have, so a simple comparison of the three groups gives the wrong answer on purpose.
3. **Work out the 6-month HbA1c change under all three drugs** for every patient: a base effect per drug, adjusted for starting HbA1c, eGFR and years with diabetes, plus random noise. The directions come from published studies (Tsapas 2020 network meta-analysis, Dennis 2022, TRIMASTER 2023).
4. **Keep only the result for the drug each patient got** (`y`). The other two are hidden, just as in real life. Only these observed columns reach the estimators (`observed_view()`).
5. **Weight change and low-sugar risk** are made the same way from a separate random stream, so adding them did not change a single primary number.

What the numbers in `params.yaml` rest on (honest count)

Section	CITED	ASSUMED-DIRECTIONAL	TEAM-SET	Total
Generator (how patients are made)	2	15	72	89
Engine settings (folds, 0.05 overlap rule, ...)	2	0	16	18
Display and context (BMI cut-offs, prevalence)	5	0	5	10
All	9	15	93	117

ASSUMED-DIRECTIONAL means the direction and rough size come from the cited study but the exact value is ours. TEAM-SET means our choice (many are plausibility limits or method settings). The loader refuses any number without a source and a status.

3. Why synthetic data is valid here, and what it does NOT prove

- In real records you never see what would have happened under the drug a patient did **not** get, so no real dataset can tell you whether a causal method got the right answer.
- With a synthetic cohort we **know** the right answer, so we can grade each method. This is how causal-inference methods are normally tested (for example the ACIC data challenges and the semi-synthetic IHDP

benchmark).

- Our result (`results/benchmark_summary.csv`, 20 cohorts × 5,000 patients): the true SGLT2i-vs-DPP-4i effect is **−0.211**; the naive comparison says **−0.370** (biased by the built-in confounding); AIPW says **−0.209** and its 95% interval contains the truth in 95% of cohorts; 9 of 9 refutation checks pass.
- **It does not prove** how much each drug lowers HbA1c in Indian patients. Those effect sizes are inputs we set from the literature, not findings.

What is real today: the safety rules (drug labels, KDIGO), the evidence and explanations (WHO, FDA), and the method. **What is synthetic:** the effect numbers on the option cards.

4. Real Indian diabetes datasets (checked 28 Sept 2026)

The key fact for the panel: **no public Indian dataset records which add-on drug each patient got together with their HbA1c months later.** That pairing is what a causal engine needs. The datasets below are real and useful for other things.

Dataset	Who / where	Size	Useful variables	Drug choice + follow-up HbA1c?	Access	How we use it
NMB-2017 (Nagarathna et al., <i>Diabetes Res Clin Pract</i> 2020)	Nationwide Indian survey of adults at high risk of type 2 diabetes	7,496 (1,986 self-declared diabetes)	age, sex, state, height, weight, waist, family history, activity, HbA1c	No	Open, CC0 1.0 , Mendeley Data doi:10.17632/twp8xw6p25.1	In the repo as a realism check (section 5)
LASI Wave 1 (IIPS, 2017–18)	Nationally representative, adults 45+	~66,000 with dried-blood-spot biomarkers	HbA1c (dried blood spot), self-reported diabetes and treatment, BMI, BP	No (one wave, no drug class)	Free request form with ID at IIPS, academic use only	Next calibration step for age, BMI and HbA1c (needs the request)
NFHS-5 (2019–21)	Nationally representative households	~700,000 women and ~100,000 men	random blood glucose, self-reported diabetes and whether on medicine, BMI	No	Registration with the DHS Program	Prevalence and context only
CARRS (Chennai + Delhi cohort)	Urban adults 20+	12,271, followed every 2 years	HbA1c, glucose, risk factors	Some follow-up, but not add-on drug choice	By request to the investigators	Possible future collaboration
ICMR-INDIAB	National prevalence study	~113,000	glucose, HbA1c subset	No	Not public (published tables only)	Already cited in <code>params.yaml</code> for context
Freedom from Diabetes clinic, Pune (PMC11937334)	18,950 T2D patients in a lifestyle programme	18,950	HbA1c, glucose-lowering medicines	Partly (medicines listed, not an add-on comparison)	"On reasonable request" to the authors	Candidate partner for real-data validation

Not Indian, and so not used: the **Pima** dataset (Pima Native American women in Arizona, no drug or outcome data); the Bangladesh and Iraq diabetes sets on Mendeley Data; the Turkish glycaemic-control set.

5. Real vs synthetic: the realism check

`python scripts/compare_cohort.py` → `docs/midsem/data/real_vs_synthetic.png` and `.csv`. It compares our cohort with NMB-2017 adults who have diabetes and HbA1c $\geq 7\%$ ($n = 1,010$), the closest real match to "on metformin, not at target".

	Synthetic (seed 0, n = 5,000)	NMB-2017, diabetes + HbA1c $\geq 7\%$ (n = 1,010)
Age, mean (SD)	55.0 (9.9)	53.5 (9.8)
BMI, mean (SD)	26.4 (4.1)	26.6 (4.3)
BMI ≥ 25 (Asian-Indian obese)	65%	60%
Female	45%	41%
HbA1c, mean (SD)	8.6 (1.0)	9.2 (1.8)

What it shows: age, BMI and sex match real Indian adults with diabetes closely. HbA1c is **lower and narrower** in our cohort than in the real data (real patients have more very high values, up to 14%). This is a concrete improvement we can make: widen the HbA1c distribution (`generator.covariates.hba1c` in `params.yaml`) and rerun the benchmark, with NMB-2017 as the cited source. Note that NMB-2017 did not ask about metformin, so it is a rough match, not an exact one.

6. How it could be used in a real hospital

1. **Data:** an extract from a hospital's records of adults on metformin who started SGLT2i, DPP-4i or a sulfonylurea: baseline details plus HbA1c 3–9 months later. Needs the hospital ethics committee (IEC), de-identification and India's DPDP Act 2023 (guide's partner hospital).
2. **Same code:** the extract has the same columns as `observed_view()`, so it replaces the synthetic cohort with no change to the estimators.
3. **Checks without a known truth:** the overlap rule (propensity < 0.05 → "insufficient evidence"), refutation tests and E-values already built, plus agreement of the average effects with published trials ("target trial emulation").
4. **Silent pilot:** doctors see the suggestion but decide as usual; record agreement and safety flags.
5. **Regulation:** use in routine care would make it software as a medical device (CDSCO). Until then it stays a research prototype; the clinician decides.

One line for the panel: "We proved the method where the truth is known; the next step is the same code on ethically approved hospital records, checked against trial results."

7. How much is implemented: about 64% of the final-year project

Weights are the team's judgement of each part's share of the whole year; each "% done" points to a file or test.

Part	Weight	Done	Evidence
Chat app: UI, guards, sign-in (CAPTCHA + MFA), patient panel, guardrails, voice, Docker	20%	~95%	392 backend + 238 frontend tests; <code>check_all</code> 43 checks
Causal engine (cohort, safety rules, propensity, IPW / AIPW / matching, DR-learner, secondary outcomes, benchmark, refutation)	25%	~90%	150 engine tests; <code>results/</code> (9 of 9 refutation checks). Missing: real-data validation
RAG (sources, hybrid search, abstention, explanations, citation checker, 60-question evaluation)	20%	~75%	33 RAG tests; <code>results/rag_eval_summary.csv</code> (recall@5 0.933, abstention 0.800, citation precision 1.000, 0 dose leaks). Missing: medical embedding model, reranker, doctor review, RSSDI source
Joining engine + RAG + LLM into the chat app	15%	~10%	Stages and contract exist; 3 of 6 stages still return "skipped"
Clinician evaluation (25 vignettes, SUS)	10%	~20%	Pack ready in <code>eval/</code> ; not yet run with doctors
Report and paper	10%	~40%	Mid-sem report done; paper not started

Part	Weight	Done	Evidence
Total	100%	≈ 64%	$0.20 \times 95 + 0.25 \times 90 + 0.20 \times 75 + 0.15 \times 10 + 0.10 \times 20 + 0.10 \times 40$

Far past the 25% asked for mid-sem: every core part works and is tested on its own. What is left: join them in the chat app, the doctor review and usability study, a better search model, the real-data step above, and the paper.

Pictures (first rows of each dataset, and the realism check)

1. Synthetic cohort: first 8 of 5,000 patients (SYNTHETIC, observed columns only)

age	female	duration_years	hba1c	egfr	bmi	ascvd	hf	hypo_history	low_income	ckd	treatment	y	weight_change_6m	hypo_6m
56	1	8.60	7.70	52	22.40	0	0	0	0	1	SU	-1.38	3.00	0
54	1	11.90	9.70	85	28.30	0	0	0	1	0	SGLT2i	-1.55	-3.48	0
61	1	0.80	9.80	76	27.80	0	0	0	0	0	SU	-1.07	2.18	0
56	1	4.20	9.70	85	22.60	0	0	0	0	0	SGLT2i	-0.37	-0.24	0
50	0	7.00	9.70	112	30.40	1	0	0	0	0	SGLT2i	-1.76	-4.66	0
59	1	2.70	8.90	102	27.90	0	0	0	0	0	DPP4i	-0.90	4.18	0
68	1	3.00	9.10	78	23.70	0	0	0	0	0	SGLT2i	-1.12	-0.75	0
64	0	8.50	9.30	66	28.90	0	0	0	0	0	SU	-1.87	0.47	1

Generated by `diacausal_engine/cohort.py` from `data/params.yaml` (seed 0); columns `dka_history`, `pancreatitis_history` and `t1d` (always 0) not shown. One row = one patient on metformin who was given one add-on drug ('treatment'). `y` = 6-month HbA1c change (percentage points); `weight_change_6m` in kg; `hypo_6m` = any low-sugar episode (1 = yes). These are the only columns the estimators see; a real hospital record would have the same ones.

2. The same patients with their hidden answers (possible only because the data is synthetic)

age	hba1c	egfr	treatment	y	y_true_SGLT2i	y_true_DPP4i	y_true_SU	e_true_SGLT2i	e_true_DPP4i	e_true_SU
56	7.70	52	SU	-1.38	-1.23	-1.27	-1.38	0.12	0.66	0.21
54	9.70	85	SGLT2i	-1.55	-1.55	-1.24	-1.53	0.32	0.15	0.53
61	9.80	76	SU	-1.07	-0.81	-0.54	-1.07	0.49	0.25	0.26
56	9.70	85	SGLT2i	-0.37	-0.37	-0.06	-0.51	0.42	0.24	0.34
50	9.70	112	SGLT2i	-1.76	-1.76	-1.28	-1.67	0.88	0.04	0.08
59	8.90	102	DPP4i	-0.90	-1.25	-0.90	-1.28	0.61	0.16	0.22
68	9.10	78	SGLT2i	-1.12	-1.12	-0.90	-1.30	0.41	0.37	0.21
64	9.30	66	SU	-1.87	-1.72	-1.55	-1.87	0.43	0.38	0.19

Shaded columns are never shown to the estimators. `y_true_X` = the 6-month HbA1c change this patient WOULD have had on drug X (`y` equals the one for the drug actually given); `e_true_X` = the true chance a doctor would pick X for this patient. Knowing these is what lets the benchmark grade each method.

3. Safety rules: first 8 of 10 (REAL: copied from drug labels and guidelines)

rule_id	arm	field	op	value	action	message	source	status
R01	SGLT2i	egfr	lt	45	EXCLUDE	Not recommended to improve glycae...	FARXIGA (dapagliflozin) US prescr...	VERIFIED
R02	SGLT2i	t1d	eq	1	EXCLUDE	Not recommended for glycaemic con...	FARXIGA US prescribing informatio...	VERIFIED
R03	SGLT2i	dka_history	eq	1	CAUTION	History of ketoacidosis: SGLT2 in...	FARXIGA US prescribing informatio...	UNVERIFIED
R04	DPP4i	egfr	lt	45	CAUTION	Renal dose adjustment required pe...	JANUVIA (sitagliptin) US prescrib...	VERIFIED
R05	DPP4i	pancreatitis_history	eq	1	CAUTION	Not studied in patients with a hi...	JANUVIA US prescribing informatio...	VERIFIED
R06	DPP4i	hf	eq	1	CAUTION	Heart failure observed with two o...	JANUVIA US prescribing informatio...	VERIFIED
R07	SU	hypo_history	eq	1	CAUTION	Sulfonylureas can cause severe hy...	Glimepiride US prescribing inform...	VERIFIED
R08	SU	age	ge	65	CAUTION	Older patients are at higher hypo...	Glimepiride US prescribing inform...	UNVERIFIED

`data/rules.csv`. Every rule has a source and a section; the engine checks them BEFORE any estimate and never estimates an excluded option.

4. Generator parameters: the drug-effect rows of data/params.yaml

parameter	value	unit	source	status
SGLT2i.base	-0.75	percentage points	TSAPAS_2020	ASSUMED-DIRECTIONAL
SGLT2i.per_hba1c_pct	-0.20	points per % above 8.5	DENNIS_2022	ASSUMED-DIRECTIONAL
SGLT2i.per_10_egfr	-0.06	points per 10 mL/min/1.73m2 above 80	TRIMASTER_2023	ASSUMED-DIRECTIONAL
SGLT2i.per_duration_year	0.00	points per year above 7	TEAM	TEAM-SET
DPP4i.base	-0.56	percentage points	TSAPAS_2020	ASSUMED-DIRECTIONAL
DPP4i.per_hba1c_pct	-0.12	points per % above 8.5	DENNIS_2022	ASSUMED-DIRECTIONAL
DPP4i.per_10_egfr	0.00	points per 10 mL/min/1.73m2 above 80	TRIMASTER_2023	ASSUMED-DIRECTIONAL
DPP4i.per_duration_year	0.00	points per year above 7	TEAM	TEAM-SET

All 117 numbers in params.yaml carry a source and a status: TEAM-SET 93, ASSUMED-DIRECTIONAL 15, CITED 9. The loader refuses any number without them.

5. Evidence sources: first 8 of 26 (REAL documents, licence-checked)

id	issuer	version	licence_as_found	bucket	date_checked
S01	WHO	2018 (ISBN 978-92-4-155028-4)	CC BY-NC-SA 3.0 IGO	cleared_ingest	2026-09-26
S02	RSSDI & Endocrine Society of ...	2020; Indian J Endocrinol Met...	CC BY-NC-SA 4.0 (journal-leve...	cleared_ingest_PENDING	2026-09-19
S03	RSSDI	2022; Int J Diabetes Dev Ctri...	(c) exclusive licence to RSSD...	cite_only	2026-09-19
S04	ICMR / Dept of Health Researc...	Undated STW PDF (2024 upload)	(c) Department of Health Rese...	cite_only_permission_requested	2026-09-19
S05	ICMR	2018	UNVERIFIED	cite_only_permission_requested	2026-09-19
S06	MoHFW, Gol	2022	UNVERIFIED	cite_only_structured	2026-09-19
S07	US FDA (Drugs@FDA / openFDA)	current labels	UNVERIFIED for reuse (manufac...	cite_only_structured	2026-09-19
S08	US FDA	2016	US federal government work	cleared_ingest	2026-09-26

RAG/sources.csv. Only rows marked cleared_ingest and checked by a named person are read into the evidence index.

6. Evidence test set: first 8 of 60 questions (written by the team)

id	question	category	expected_source	doctor_review
G01	Which medicine does WHO recommend adding when metformin alo...	answerable	S01	yes
G02	Which sulfonylurea should be avoided in people aged 60 and ...	answerable	S01	yes
G03	Which sulfonylurea is preferred when hypoglycaemia is a con...	answerable	S01	yes
G04	Do DPP-4 inhibitors lower HbA1c as much as sulfonylureas wh...	answerable	S01	yes
G05	Is severe hypoglycaemia less likely with SGLT-2 inhibitors ...	answerable	S01	yes
G06	Which add-on medicines were associated with weight loss in ...	answerable	S01	no
G07	Why did the WHO guideline group not prefer the newer oral m...	answerable	S01	yes
G08	What did the guideline group say about the cardiovascular b...	answerable	S01	no

eval/rag_gold.csv: 45 answerable, 10 out of scope (must abstain), 5 dose questions (must refuse). 20 are marked for a doctor's review.

7. Real Indian data: NMB-2017, first 8 of 7,496 adults (REAL, CC0, reference only)

patient_id	state	district	age	gender	height_cm	weight_kg	waist_cm	diabetes_self_declared	diabetes_father	diabetes_mother	hba1c
0	Rajasthan	Ajmer	35	Female	157	47	70	Normal	0	0	4.90
1	Orissa	Balasore	38	Female	145	65	81	Normal	1	0	5.30
4	Rajasthan	Ajmer	42	Female	159	68	102	Normal	1	1	5.00
7	Orissa	Balasore	37	Female	142	44	76	Normal	1	0	5.40
8	Orissa	Balasore	47	Male	160	60	87	Normal	1	0	12.60
10	Karnataka	Bangalore	59	Female	158	57	83	Normal	0	0	6.30
13	Orissa	Balasore	30	Female	151	84	72	Normal	1	0	6.20
14	Orissa	Balasore	37	Female	151	60	81	Normal	1	0	4.90

Nagarathna et al. 2020, Diabetes Res Clin Pract; Mendeley Data doi:10.17632/tpw8xw6p25.1 (CC0 1.0). Nationwide survey of adults at high risk of type 2 diabetes. No drug and no follow-up HbA1c, so it cannot train the engine; used to check the synthetic cohort looks real. Activity columns and zone not shown.

Synthetic cohort vs real Indian adults with diabetes and HbA1c $\geq 7\%$ (n = 1010)

